

CHAPTER II.5, WEEK 7

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5.9. Let S be a graded ring, generated by S_1 as an S_0 -algebra, M a graded S -module, and $X = \text{Proj } S$.

- (a) Show there is a natural homomorphism $\alpha : M \rightarrow \Gamma_*(\widetilde{M})$.
- (b) Assume now that $S_0 = A$ is a finitely generated k -algebra for some field k , that S_1 is a finitely generated A -module, and that M is a finitely generated S -module. Show that the map α is an isomorphism in all large enough degrees, i.e., there is a $d_0 \in \mathbb{Z}$ such that for all $d \geq d_0$, $\alpha_d : M_d \rightarrow \Gamma(X, \widetilde{M}(d))$ is an isomorphism. [Hint: Use the methods of the proof of (5.19).]
- (c) With the same hypothesis, we define an equivalence relation $\approx$ on graded S -modules by saying $M \approx M'$ if there is an integer d such that $M_{\geq d} \cong M'_{\geq d}$. Here, $M_{\geq d} = \bigoplus_{n \geq d} M_n$. We say that a graded S -module is *quasi-finitely generated* if it is equivalent to a finitely generated module. Now show that the functors $\widetilde{}$ and Γ_* induces an equivalence of categories between the category of quasi-finitely generated graded S -modules modulo the equivalence relation $\approx$, and the category of coherent $\mathcal{O}_X$ -modules.

Proof. (a) Fix generators $\{x_i\}_{i \in I} \subset S_1$ of S as an S_0 -algebra. This gives us an affine open cover $\{D_+(x_i) \cong \text{Spec } S_{(x_i)}\}_{i \in I}$ of $\text{Proj } S$ such that

$$\begin{aligned} \widetilde{M}|_{D_+(x_i)}(n) &\cong \widetilde{M(n)}|_{D_+(x_i)} \\ &\cong (M(n)_{(x_i)})^\sim \\ &\cong ((M_{x_i})_n)^\sim \end{aligned}$$

since restriction commute with tensor products, and $x_i \in S_1$ implies

$$(M \otimes_S N)_{(x_i)} = M_{(x_i)} \otimes_{S_{(x_i)}} N_{(x_i)}.$$

Then

$$\begin{aligned} \Gamma_*(\widetilde{M}|_{D_+(x_i)}) &\cong \bigoplus_{n \in \mathbb{Z}} \Gamma(D_+(x_i), \widetilde{M}|_{D_+(x_i)}(n)) \\ &\cong \bigoplus_{n \in \mathbb{Z}} \Gamma(\text{Spec } S_{(x_i)}, ((M_{x_i})_n)^\sim) \\ &\cong \bigoplus_{n \in \mathbb{Z}} (M_{x_i})_n \\ &\cong M_{x_i} \end{aligned}$$

supplies us with a natural map $\alpha_i : M \hookrightarrow M_{x_i} \xrightarrow{\sim} \Gamma_*(\widetilde{M}|_{D_+(x_i)})$ for each i . Using a similar argument to the above, we define maps $\alpha_{ij} : M \hookrightarrow M_{x_i x_j} \xrightarrow{\sim} \Gamma_*(\widetilde{M}|_{D_+(x_i x_j)})$ and show that $\rho_{D_+(x_i x_j)} \circ \alpha_i = \alpha_{ij} = \rho_{D_+(x_i x_j)} \circ \alpha_j$.

Fixing $m \in M$, the $\alpha_i(m)_n \in \Gamma(D_+(x_i), \widetilde{M}|_{D_+(x_i)}(n))$ glue together to a section $\alpha(m)_n \in \Gamma(X, \widetilde{M})$. Only finitely many $\alpha_i(m)_n$ are nonzero for each i , so only finitely many $\alpha(m)_n$ are nonzero, whence $\alpha(m) = \sum_n \alpha(m)_n \in \Gamma_*(\widetilde{M})$ is well-defined. We take $\alpha : m \mapsto \alpha(m)$ to be our natural map.

(b)

□

5.10. Let A be a ring, $S = A[x_0, \dots, x_r]$, and $X = \text{Proj } S$. We have seen that a homogeneous ideal I in S defines a closed subscheme of X (Ex. 3.12), and that conversely every closed subscheme of X arises in this way (5.16).

(a) For any homogeneous ideal $I \subset S$, we define the saturation $\bar{I}$ of I to be

$$\bar{I} := \{s \in S \mid \forall 0 \leq i \leq r \ (\exists n > 0 \ (x_i^n s \in I))\}$$

We say that I is *saturated* if $I = \bar{I}$. Show that $\bar{I}$ is a homogeneous ideal of S .

- (b) Two homogeneous ideals I_1 and I_2 of S define the same closed subscheme of X iff they have the same saturation.
- (c) If Y is any closed subscheme of X , then the ideal $\Gamma_*(\mathcal{I}_Y)$ is saturated. Hence, it is the largest homogeneous ideal defining the subscheme Y .
- (d) There is a 1-1 correspondence between saturated ideals of S and closed subschemes of X .

Proof. (a) If $s \in \bar{I}$, then we can write $s = \sum_d s_d$ as the sum of its homogeneous parts, so that $x_i^n s = \sum_d x_i^n s_d \in I$ and I homogeneous implies $x_i^n s_d \in I$ for each d . We conclude that $s_d \in \bar{I}$ for all d , whence $\bar{I}$ is homogeneous.

(b) Fix homogeneous ideals $I_1, I_2 \subseteq S$. We have an affine cover $D_+(x_i) \cong \text{Spec } S_{(x_i)}$ of X , so it suffices to show that

$$\frac{S_{(x_i)}}{(I_1)_{(x_i)}} \cong \widetilde{S/I_1}|_{D_+(x_i)} \cong \widetilde{S/I_2}|_{D_+(x_i)} \cong \frac{S_{(x_i)}}{(I_2)_{(x_i)}}$$

for all $0 \leq i \leq r$ iff $\bar{I}_1 = \bar{I}_2$. This occurs iff the kernels of the respective projection maps are the same, i.e. $(I_1)_{(x_i)} = (I_2)_{(x_i)}$. If $\bar{I}_1 = \bar{I}_2$, then $\exists n > 0$ with $x_i^n s \in I_1$ iff $\exists n > 0$ with $x_i^n s \in I_2$.

(c)

□

5.11. Let S and T be two graded rings with $S_0 = T_0 = A$. We define the *Cartesian product* $S \times_A T$ to be the graded ring $\bigoplus_{d \geq 0} S_d \otimes_A T_d$. If $X = \text{Proj } S$ and $Y = \text{Proj } T$, show that $\text{Proj}(S \times_A T) \cong X \times_A Y$, and show that the sheaf $\mathcal{O}(1)$ on $\text{Proj}(S \times_A T)$ is isomorphic to the sheaf $p_1^*(\mathcal{O}_X(1)) \otimes p_2^*(\mathcal{O}_Y(1))$ on $X \times_A Y$.

The Cartesian product of rings is related to the Segre embedding of projective spaces in the following way. If $x_0, \dots, x_r$ is a set of generators for S_1 over A , corresponding to a projective embedding $X \hookrightarrow \mathbb{P}_A^r$, and if $y_0, \dots, y_s$ is a set of generators for T_1 , corresponding to a projective embedding $Y \hookrightarrow \mathbb{P}_A^s$, then $x_i \otimes y_j$ is a set of generators for $(S \times_A T)_1$, and hence defines a projective embedding $\text{Proj}(S \times_A T) \hookrightarrow \mathbb{P}_A^N$, with $N = rs + r + s + 1$. This is just the image of $X \times Y \subset \mathbb{P}^r \times \mathbb{P}^s$ in its Segre embedding.

Proof.

□

- 5.12. (a) Let X be a scheme over a scheme Y , and let $\mathcal{L}, \mathcal{M}$ be two very ample invertible sheaves on X . Show that $\mathcal{L} \otimes \mathcal{M}$ is also very ample. [Hint: Use a Segre embedding.]
- (b) Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two morphisms of schemes. Let $\mathcal{L}$ be a very ample invertible sheaf on X relative to Y , and let $\mathcal{M}$ be a very ample invertible sheaf on Y relative to Z . Show that $\mathcal{L} \otimes f^*\mathcal{M}$ is a very ample invertible sheaf on X relative to Z .

Proof.

□

- 5.13. Let S be a graded ring, generated by S_1 as an S_0 -algebra. For any integer $d > 0$, let $S^{(d)}$ be the graded ring $\bigoplus_{n \geq 0} S_n^{(d)}$, where $S_n^{(d)} = S_{nd}$. Let $X = \text{Proj } S$. Show that $\text{Proj } S^{(d)} \cong X$, and that the sheaf $\mathcal{O}(1)$ on $\text{Proj } S^{(d)}$ corresponds via this isomorphism to $\mathcal{O}_X(d)$.

This construction is related to the d -uple *embedding* in the following way. Denote $A = S_0$. If $x_0, \dots, x_r$ is a set of generators for S_1 , corresponding to an embedding $X \hookrightarrow \mathbb{P}_A^r$, then the set of monomials of degree d in the x_i is a set of generators for $S_1^{(d)} = D_d$. These define a projective embedding of $\text{Proj } S^{(d)}$ which is none other than the image of X under the d -uple embedding of $\mathbb{P}_A^r$.

- 5.14. Let A be a ring, and let X be a closed subscheme of $\mathbb{P}_A^r$. We define the homogeneous coordinate ring $S(X)$ of X for the given embedding to be $A[x_0, \dots, x_r]/I$, where I is the ideal $\Gamma_*(\mathcal{I}_X)$ constructed in the proof of (5.14). (Of course, if A is a field and X a variety, this coincides with the usual definition!) Recall that a scheme X is *normal* if its local rings are integrally closed domains. A closed subscheme $X \subset \mathbb{P}_A^r$ is *projectively normal* for the given embedding, if the homogeneous coordinate ring $S(X)$ is an integrally closed domain. Now assume that A is a finitely generated k -algebra for some field k , and that X is a connected, normal closed subscheme of $\mathbb{P}_A^r$. Show that for some $d > 0$, the d -uple embedding of X is projectively normal, as follows.

- (a) Let S be the homogeneous coordinate ring of X , and let $S' = \bigoplus_{n \geq 0} \Gamma(X, \mathcal{O}_X(n))$. Show that S is a domain, and that S' is its integral closure. [Hint: First show that X is integral. Then regard S' as the global sections of the sheaf of rings $\mathcal{S} = \bigoplus_{n \geq 0} \mathcal{O}_X(n)$ on X , and show that $\mathcal{S}$ is a sheaf of integrally closed domains.]
- (b) Use (Ex. 5.9) to show that $S_d = S'_d$ for all sufficiently large d .
- (c) Show that $S^{(d)}$ is integrally closed for sufficiently large d , and hence conclude that the d -uple embedding of X is projectively normal.
- (d) As a corollary of (a), show that a closed subscheme $X \subset \mathbb{P}_A^r$ is projectively normal if and only if it is normal, and for every $n \geq 0$ the natural map $\Gamma(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}(n)) \rightarrow \Gamma(X, \mathcal{O}_X(n))$ is surjective.